

Exact Homological Invariants for Closed-Neighborhood Ideals of Cubes of Broom Graphs

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Abstract

Let $B(n, m)$ be the broom graph formed from a path $y_1, \dots, y_m$ by attaching n leaves to y_1 , and let $I_{n,m}$ be the closed-neighborhood ideal of its third graph power. Over an arbitrary field, we determine the projective dimension, Castelnuovo–Mumford regularity, and height of $S/I_{n,m}$ for every $n, m \geq 2$. The formulas are uniform, including all small values of m . The proof uses minimal multigraded mapping cones and two interval-ideal recurrences; a disjoint-support packing and an explicit cover give the height. We also classify the Cohen–Macaulay cases.

Keywords. Closed-neighborhood ideal; graph power; broom graph; projective dimension; Castelnuovo–Mumford regularity; Cohen–Macaulay ring.

1 Statement of the result

Fix an arbitrary field K . For integers $n, m \geq 2$, let $B(n, m)$ have vertex set

$$\{x_1, \dots, x_n, y_1, \dots, y_m\}$$

and edge set

$$\{\{x_i, y_1\} : 1 \leq i \leq n\} \cup \{\{y_j, y_{j+1}\} : 1 \leq j < m\}.$$

Its third graph power joins two distinct vertices when their distance in $B(n, m)$ is at most three. In the standard graded ring

$$S = K[x_1, \dots, x_n, y_1, \dots, y_m],$$

write

$$I_{n,m} = NI(B(n, m)^3) = \left(\prod_{w \in N_{B(n, m)^3}[v]} w : v \in V(B(n, m)) \right).$$

The square, or radius-two, version for broom and double-broom graphs is studied by A. Olteanu and O. Olteanu [1]; the present note treats the adjacent radius-three family. No claim of bibliographic priority is made here.

Theorem 1. *For every field K and all integers $n, m \geq 2$,*

$$\mathrm{pd}_S(S/I_{n,m}) = \left\lfloor \frac{m}{4} \right\rfloor + 1,$$

$$\mathrm{reg}_S(S/I_{n,m}) = n + m - 1 - \left\lfloor \frac{m}{4} \right\rfloor,$$

$$\mathrm{ht}(I_{n,m}) = \left\lfloor \frac{m}{7} \right\rfloor + 1.$$

There are no exceptional small- m values outside these formulas. Moreover, $S/I_{n,m}$ is Cohen–Macaulay if and only if

$$m \in \{2, 3, 7\}.$$

This classification is independent of $n \geq 2$.

We prove the theorem in Sections 2–6. All arguments are monomial and hence characteristic-free.

2 Minimal generators

Put $X = x_1 \cdots x_n$, and for $a \leq b$ write $Y[a, b] = y_a \cdots y_b$. Direct distance counting gives

$$q_{x_i} = XY[1, \min\{m, 3\}]$$

for every leaf, whereas the monomial attached to y_j is

$$q_{y_j} = \begin{cases} XY[1, \min\{m, j+3\}], & 1 \leq j \leq 3, \\ Y[j-3, \min\{m, j+3\}], & 4 \leq j \leq m. \end{cases}$$

After duplicates and proper multiples are removed, this becomes

$$G(I_{n,m}) = \begin{cases} \{XY[1, m]\}, & 2 \leq m \leq 3, \\ \{XY[1, 3]\} \cup \{Y[a, a+6] : 1 \leq a \leq m-7\} \cup \{Y[m-3, m]\}, & m \geq 4. \end{cases} \quad (1)$$

The middle set is empty for $m \leq 7$. Indeed, $XY[1, 3]$ divides the three generators centered at y_1, y_2, y_3 . At the other end of the path, $Y[m-3, m]$ divides every longer right-truncated ball. The remaining displayed supports contain none of one another. This also accounts directly for all endpoints $m = 2, \dots, 7$.

3 A minimal mapping-cone lemma

The following observation will ensure that each recurrence below is exact, with no cancellation in the mapping cone.

Lemma 2. *Let $R = K[u, z_1, \dots, z_s]$, let A be a monomial ideal not involving u , and let $f = uv$, where v is a monomial in the z -variables. Then the exact sequence*

$$0 \longrightarrow R/(A : v)(-\deg f) \xrightarrow{f} R/A \longrightarrow R/(A + (f)) \longrightarrow 0 \quad (2)$$

has a minimal mapping-cone resolution. Consequently,

$$\mathrm{pd} R/(A + (f)) = \max\{\mathrm{pd} R/A, \mathrm{pd} R/(A : v) + 1\}, \quad (3)$$

$$\mathrm{reg} R/(A + (f)) = \max\{\mathrm{reg} R/A, \mathrm{reg} R/(A : v) + \deg f - 1\}. \quad (4)$$

The same conclusion holds when u is replaced by a product of variables that do not occur in A .

Proof. Since u does not occur in A , one has $(A : f) = (A : v)$, which proves exactness on the left of (2). Extend minimal multigraded resolutions of A and $(A : v)$ from $K[z_1, \dots, z_s]$ to R . After the multidegree shift by f , every nonzero entry of a homogeneous comparison map lifting multiplication by f has positive u -degree. Such an entry lies in the homogeneous maximal ideal. The two original differentials are minimal as well, so the mapping cone is minimal. Its module in homological degree i is the direct sum of the degree- i module resolving R/A and the shifted degree- $(i-1)$ module resolving $R/(A : v)$. This gives (3) and (4). The same multidegree argument applies to a product of new variables. $\square$

4 Two interval ideals

For $t \geq 0$, define $L_t \subseteq K[z_1, \dots, z_t]$ by

$$L_t = \begin{cases} 0, & 0 \leq t \leq 3, \\ (z_{t-3}z_{t-2}z_{t-1}z_t), & 4 \leq t \leq 7, \\ (z_a \cdots z_{a+6} : 1 \leq a \leq t-7) + (z_{t-3}z_{t-2}z_{t-1}z_t), & t \geq 8. \end{cases} \quad (5)$$

Set

$$p_t = \text{pd}(K[z_1, \dots, z_t]/L_t), \quad r_t = \text{reg}(K[z_1, \dots, z_t]/L_t).$$

For $t \geq 8$, view L_{t-1} as an ideal on $z_2, \dots, z_t$. Generator-by-generator monomial division gives

$$L_t = (z_1 \cdots z_7) + L_{t-1}, \quad (L_{t-1} : z_1 \cdots z_7) = (z_8) + L_{t-8}, \quad (6)$$

where the copy of L_{t-8} uses $z_9, \dots, z_t$. For the colon identity, the windows beginning at $z_2, \dots, z_8$ yield multiples of z_8 , while the later windows and the terminal four-window give exactly L_{t-8} . This reasoning includes $8 \leq t \leq 11$, when $L_{t-8} = 0$ and the terminal quotient is a multiple of z_8 .

The quotient by the colon ideal in (6) has projective dimension $p_{t-8} + 1$ and regularity r_{t-8} . Lemma 2, using z_1 as the new variable, therefore gives

$$p_t = \max\{p_{t-1}, p_{t-8} + 2\}, \quad r_t = \max\{r_{t-1}, r_{t-8} + 6\}. \quad (7)$$

The initial values are $(p_t, r_t) = (0, 0)$ for $0 \leq t \leq 3$ and $(p_t, r_t) = (1, 3)$ for $4 \leq t \leq 7$. Induction in (7) now yields

$$p_t = \left\lfloor \frac{t}{4} \right\rfloor, \quad r_t = 3 \left\lfloor \frac{t}{4} \right\rfloor \quad (t \geq 0). \quad (8)$$

Indeed, the second terms in the two maxima are respectively $\lfloor t/4 \rfloor$ and $3\lfloor t/4 \rfloor$, while the first terms are no larger.

Next, for $t \geq 1$, let $Q_t \subseteq K[z_1, \dots, z_t]$ have the minimal supports of the closed radius-three balls of a path. Thus $Q_t = (z_1 \cdots z_t)$ for $1 \leq t \leq 4$, and for $t \geq 5$,

$$Q_t = (z_1 z_2 z_3 z_4, z_a \cdots z_{a+6} \ (2 \leq a \leq t-7), z_{t-3} z_{t-2} z_{t-1} z_t), \quad (9)$$

where the middle list is empty for $t \leq 8$. Shifting the variables of L_{t-1} , one has for every $t \geq 5$

$$Q_t = (z_1 z_2 z_3 z_4) + L_{t-1}, \quad (L_{t-1} : z_1 z_2 z_3 z_4) = Q_{t-4}, \quad (10)$$

where Q_{t-4} uses $z_5, \dots, z_t$. For $5 \leq t \leq 8$, this follows by dividing the terminal four-window. Thereafter, the first surviving seven-window yields the left four-window $z_5 z_6 z_7 z_8$, the next three quotients are its multiples, and the remaining generators are exactly those of Q_{t-4} .

Let a_t and b_t denote the projective dimension and regularity of the quotient by Q_t . From Lemma 2, (8), and (10),

$$a_t = \max\{p_{t-1}, a_{t-4} + 1\}, \quad b_t = \max\{r_{t-1}, b_{t-4} + 3\}.$$

The principal cases $1 \leq t \leq 4$ have $a_t = 1$ and $b_t = t - 1$. An induction, residue class by residue class modulo four, gives

$$a_t = \left\lceil \frac{t}{4} \right\rceil, \quad b_t = t - \left\lfloor \frac{t}{4} \right\rfloor \quad (t \geq 1). \quad (11)$$

For completeness, the second entries in the two maxima are exactly the right-hand sides of (11); the first entries are no larger.

5 Projective dimension and regularity

Suppose first that $m \geq 4$. By (1),

$$I_{n,m} = (XY[1, 3]) + L_m, \quad (12)$$

where $L_m \subseteq K[y_1, \dots, y_m]$. Dividing and minimizing its generators gives

$$(L_m : Y[1, 3]) = Q_{m-3} \quad (13)$$

on $y_4, \dots, y_m$. At $4 \leq m \leq 7$, the terminal four-window divides to the full product on these $m - 3$ variables. For $m \geq 8$, the first seven-window gives $Y[4, 7]$, the next three quotients are multiples of it, and the later windows together with the terminal generator give the rest of Q_{m-3} .

Apply Lemma 2 to (12). The variables $x_1, \dots, x_n$ are new for L_m , and $\deg(XY[1, 3]) = n + 3$. Equations (8), (11), and (13) imply

$$\begin{aligned} \text{pd } S/I_{n,m} &= \max \left\{ \left\lfloor \frac{m}{4} \right\rfloor, \left\lceil \frac{m-3}{4} \right\rceil + 1 \right\} = \left\lfloor \frac{m}{4} \right\rfloor + 1, \\ \text{reg } S/I_{n,m} &= \max \left\{ 3 \left\lfloor \frac{m}{4} \right\rfloor, n + 2 + (m - 3) - \left\lceil \frac{m-3}{4} \right\rceil \right\} \\ &= n + m - 1 - \left\lfloor \frac{m}{4} \right\rfloor. \end{aligned}$$

Here $\lceil (m-3)/4 \rceil = \lfloor m/4 \rfloor$. If $m = 4c + s$ with $0 \leq s \leq 3$, the second term in the regularity maximum exceeds the first by $n + s - 1 > 0$. When $m = 2, 3$, equation (1) is principal of degree $n + m$, so the same two formulas hold. Thus no exceptional small values remain.

6 Height and Cohen–Macaulayness

Since $I_{n,m}$ is squarefree, its height is the minimum size of a vertex cover of the supports in (1). It is one for $m = 2, 3$. For $4 \leq m \leq 6$, the single vertex y_{m-3} meets both displayed generators, so the height is again one.

Now let $m \geq 7$, write $m = 7q + r$ with $0 \leq r \leq 6$, and hence $q = \lfloor m/7 \rfloor$. The following $q + 1$ generator supports are pairwise disjoint:

$$\text{supp}(XY[1, 3]), \quad \text{supp}(Y[4 + 7i, 10 + 7i]) \quad (0 \leq i \leq q - 2), \quad \text{supp}(Y[m - 3, m]). \quad (14)$$

The middle list is empty when $q = 1$. Its last interval, when present, ends at $7q - 4$, before the terminal interval begins. Thus every cover has at least $q + 1$ vertices. Conversely,

$$\{y_{3+7i} : 0 \leq i \leq q - 1\} \cup \{y_{m-3}\} \quad (15)$$

is a set of $q + 1$ distinct vertices meeting every generator. It meets the leaf and terminal supports, and no interval of seven consecutive path vertices avoids it: successive marked indices are seven apart, while the last gap is $r + 1 \leq 7$. Therefore

$$\text{ht}(I_{n,m}) = \left\lfloor \frac{m}{7} \right\rfloor + 1$$

for all $m \geq 2$.

Finally, Auslander–Buchsbaum gives

$$S/I_{n,m} \text{ Cohen–Macaulay} \iff \text{pd } S/I_{n,m} = \text{ht}(I_{n,m}).$$

By the formulas already proved, this is equivalent to

$$\left\lfloor \frac{m}{4} \right\rfloor = \left\lfloor \frac{m}{7} \right\rfloor.$$

The common value zero gives $m = 2, 3$, and the common value one gives only $m = 7$. For a common value $k \geq 2$, the intervals $[4k, 4k + 3]$ and $[7k, 7k + 6]$ are disjoint. This proves the final assertion of Theorem 1.

Computational and AI-assistance disclosure

Exact finite calculations, not used in the proof, reconstructed the distance-three balls by breadth-first search, minimized their supports, and computed minimal resolutions with SINGULAR 4.4.1. They covered $n \in \{2, 3\}$, $2 \leq m \leq 24$, over characteristic zero and characteristic two, and found no discrepancy in the generators, invariants, heights, or predicted dependence on n . The heights were also checked by exhaustive hitting-set search. As a methodological comparison only, exact computation with independently reconstructible certificates has been used in a different problem on graph-associated monomial algebras [2]; that work does not study closed-neighborhood ideals or graph powers.

An AI system assisted with proof organization, checking the displayed monomial divisions and recurrences, finite exact verification, and manuscript preparation. The quantified theorem does not rely on finite computation: its proof is the mapping-cone and covering argument given above.

References

- [1] A. Olteanu and O. Olteanu, *On the closed neighborhood ideal of the square of broom and double broom graphs*, arXiv:2609.04831v1 (2026), <https://arxiv.org/abs/2609.04831>.
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